

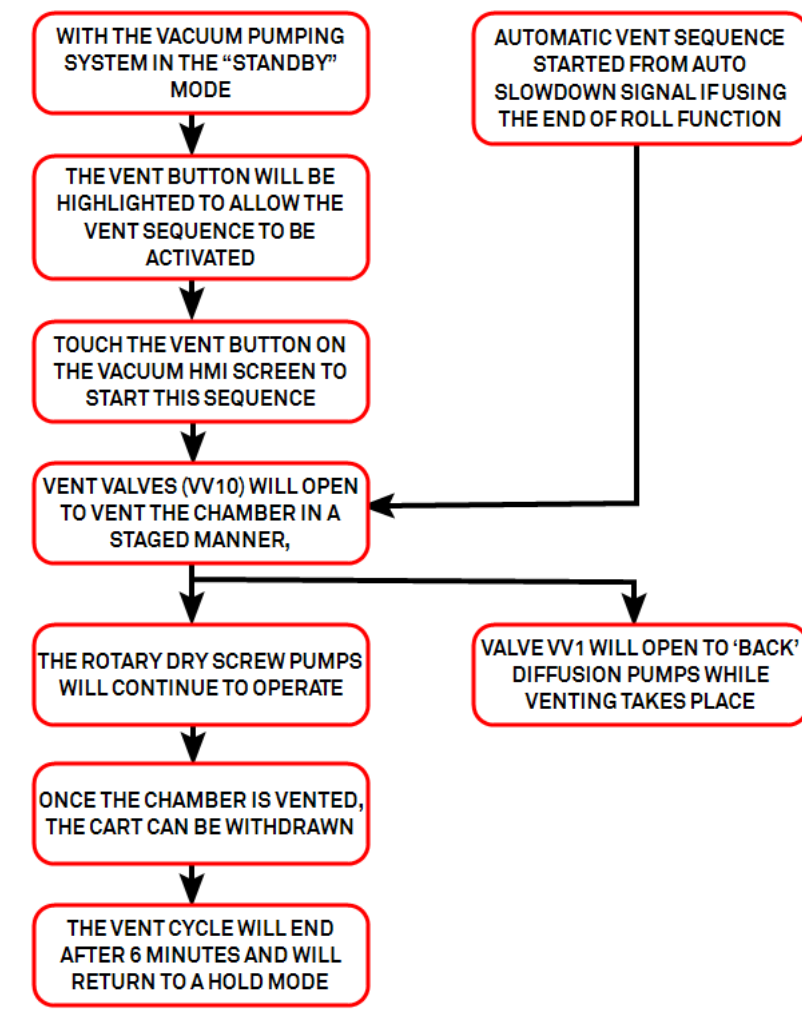


Figure 7.2-3 Chamber Vent Sequence

7.3 Winding System Chucking/Unchucking Sequences

The following explanation is provided to guide the user in the process of installing and removing rolls of substrate from the winding system. This explanation assumes that there are no rolls/shafts in the winding mechanism.

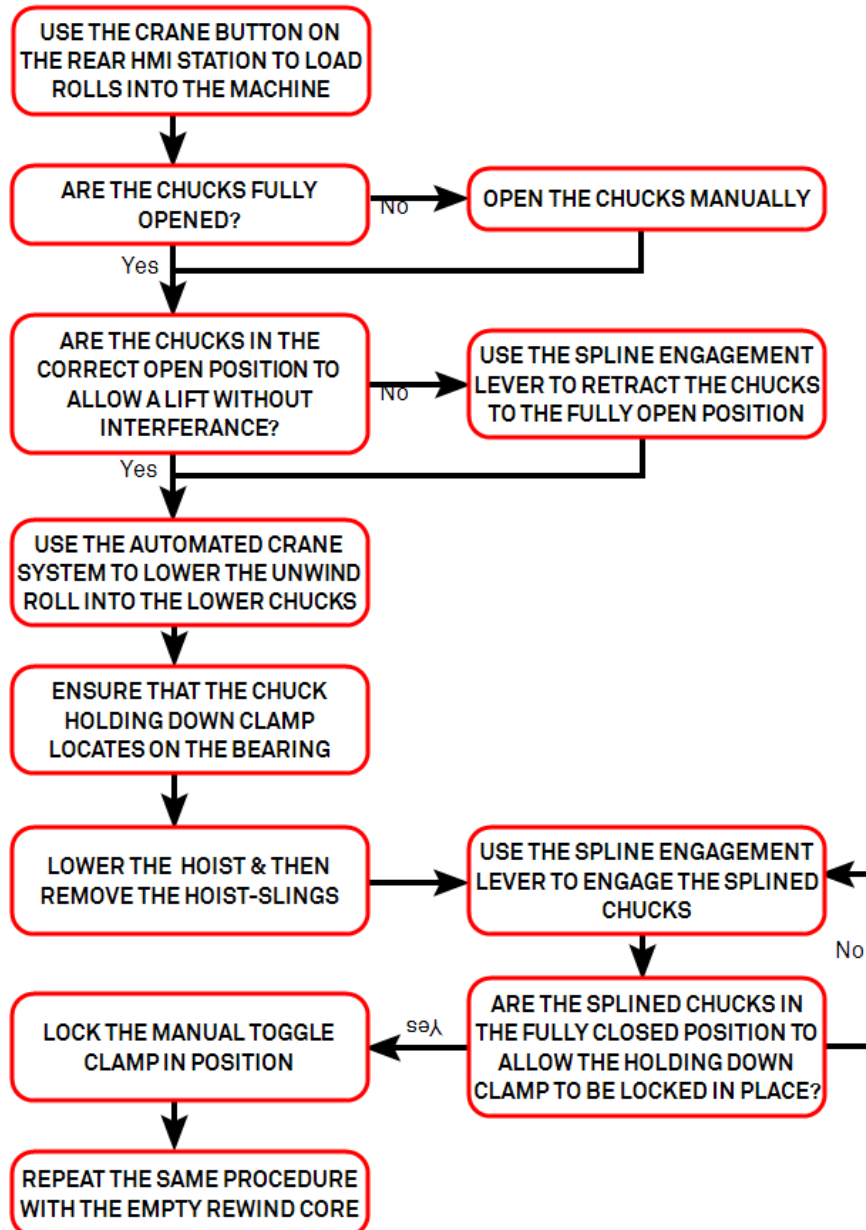


Figure 7.3-1 Chucking flow chart

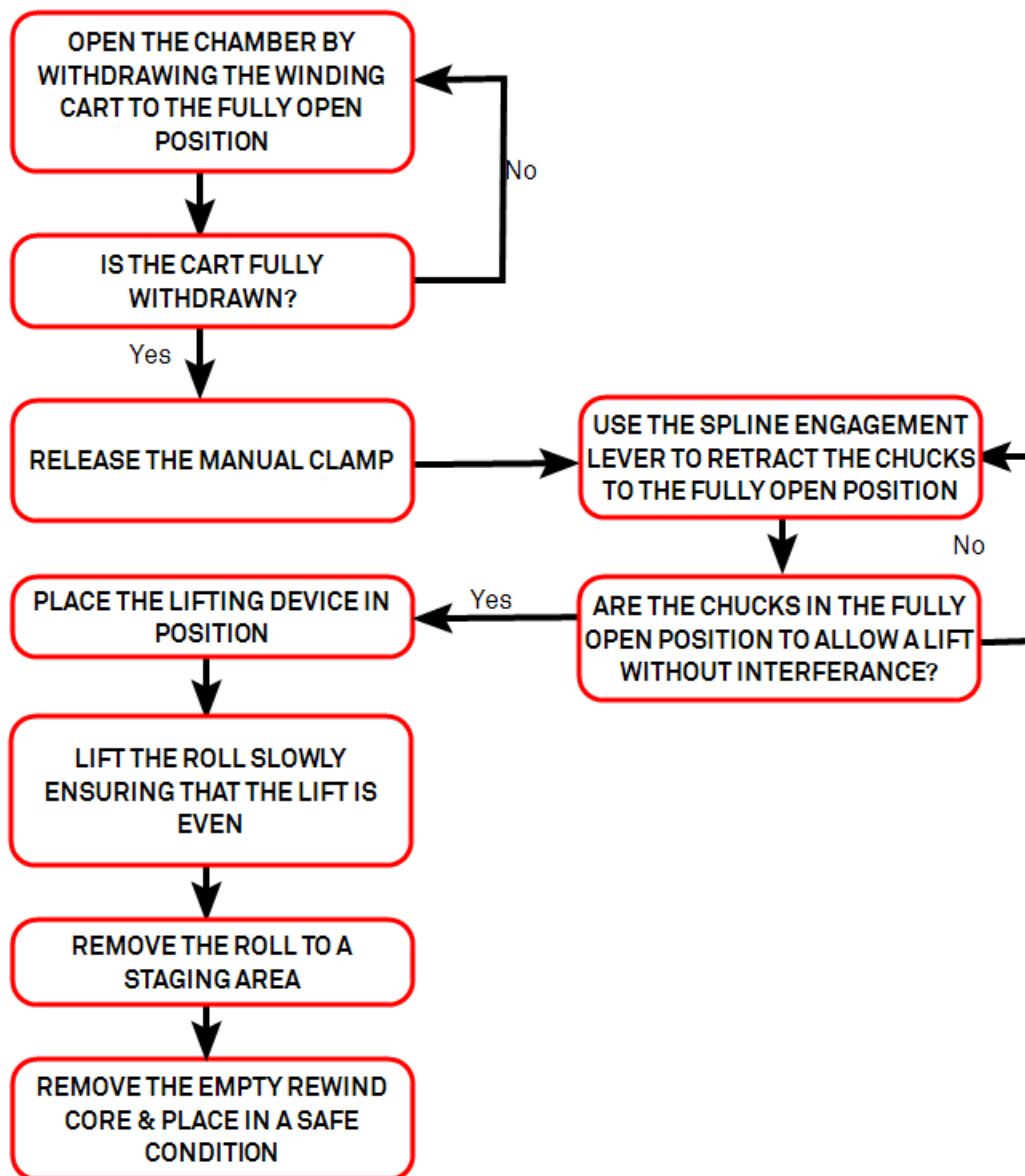


Figure 7.3-2 Unchucking flow chart rewind roll

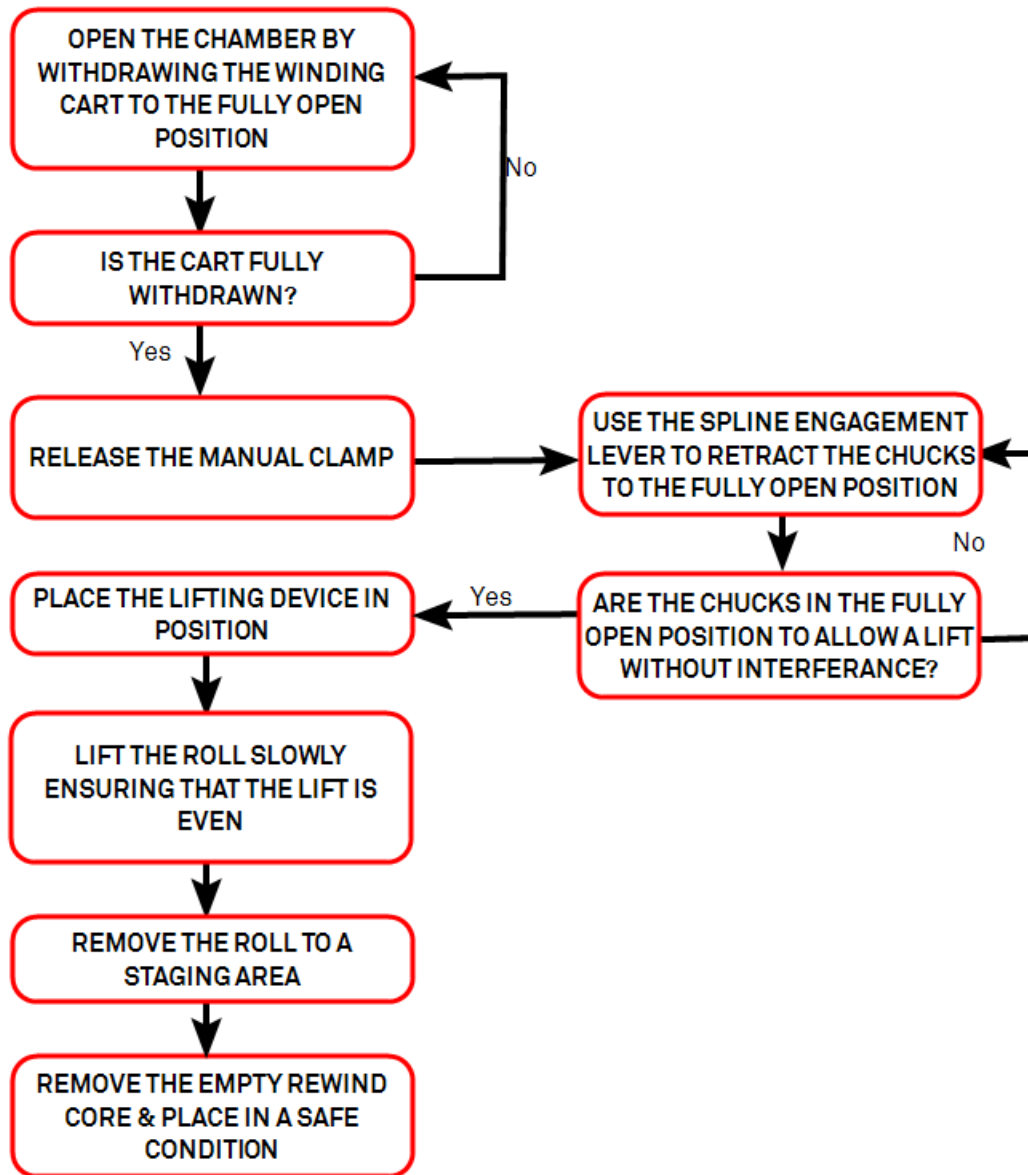


Figure 7.3-3 Unchucking flow chart unwind roll

7.4 Operational Sequence of Standard Aluminium Metallizing Process

Rather than trying to explain the sequence of operating the machine for metallizing with in-depth written descriptions, flow charts are provided, to help the operator of the machine follow the correct operational sequences.

It has been found that operators of this type of equipment prefer this type of information, while operating the machine.

At relevant sections throughout the charts, reference is made to other sections in this manual (section numbers), where further information is provided about any relevant subject.

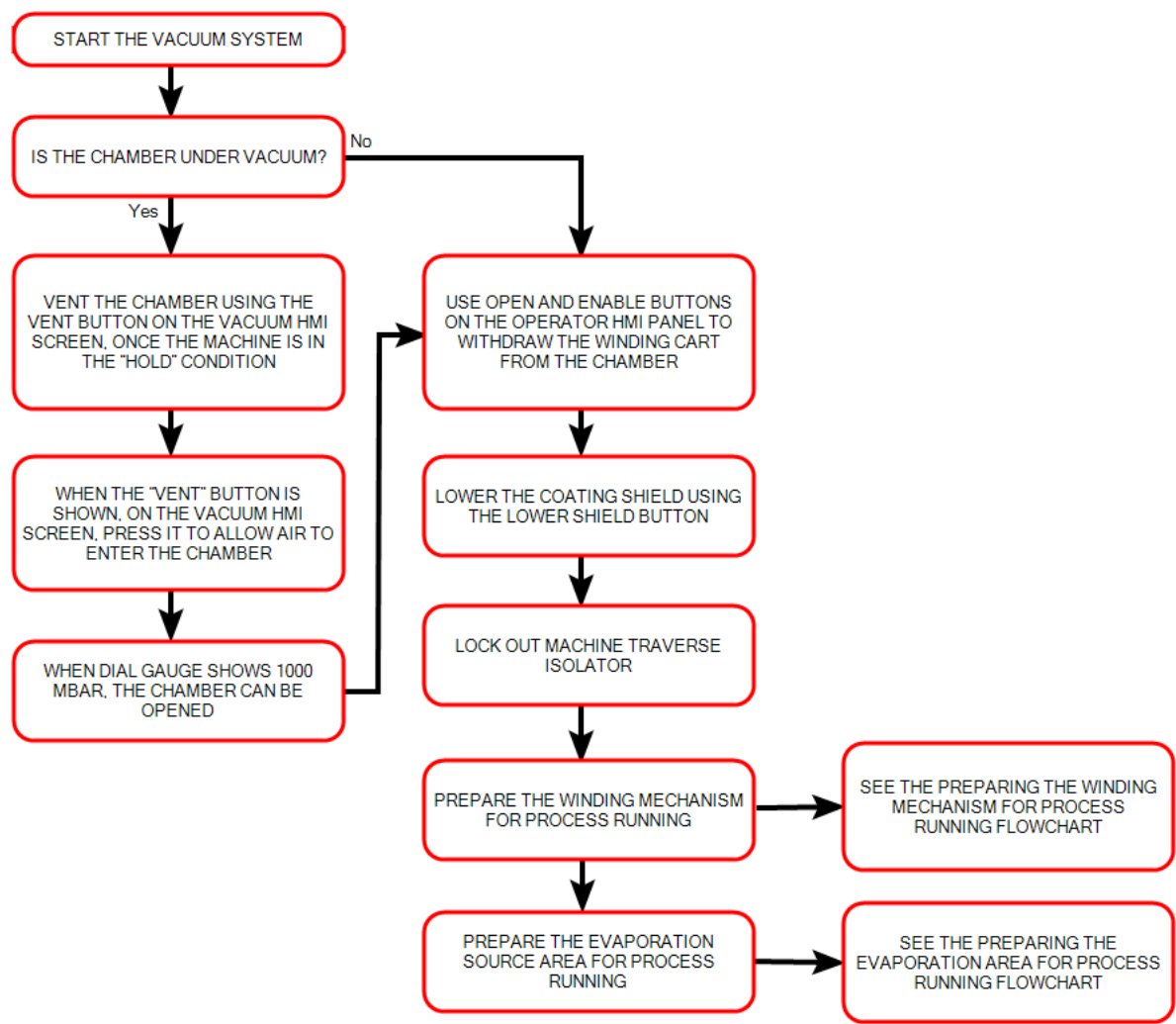


Figure 7.4-1 Machine Operation Flow Chart

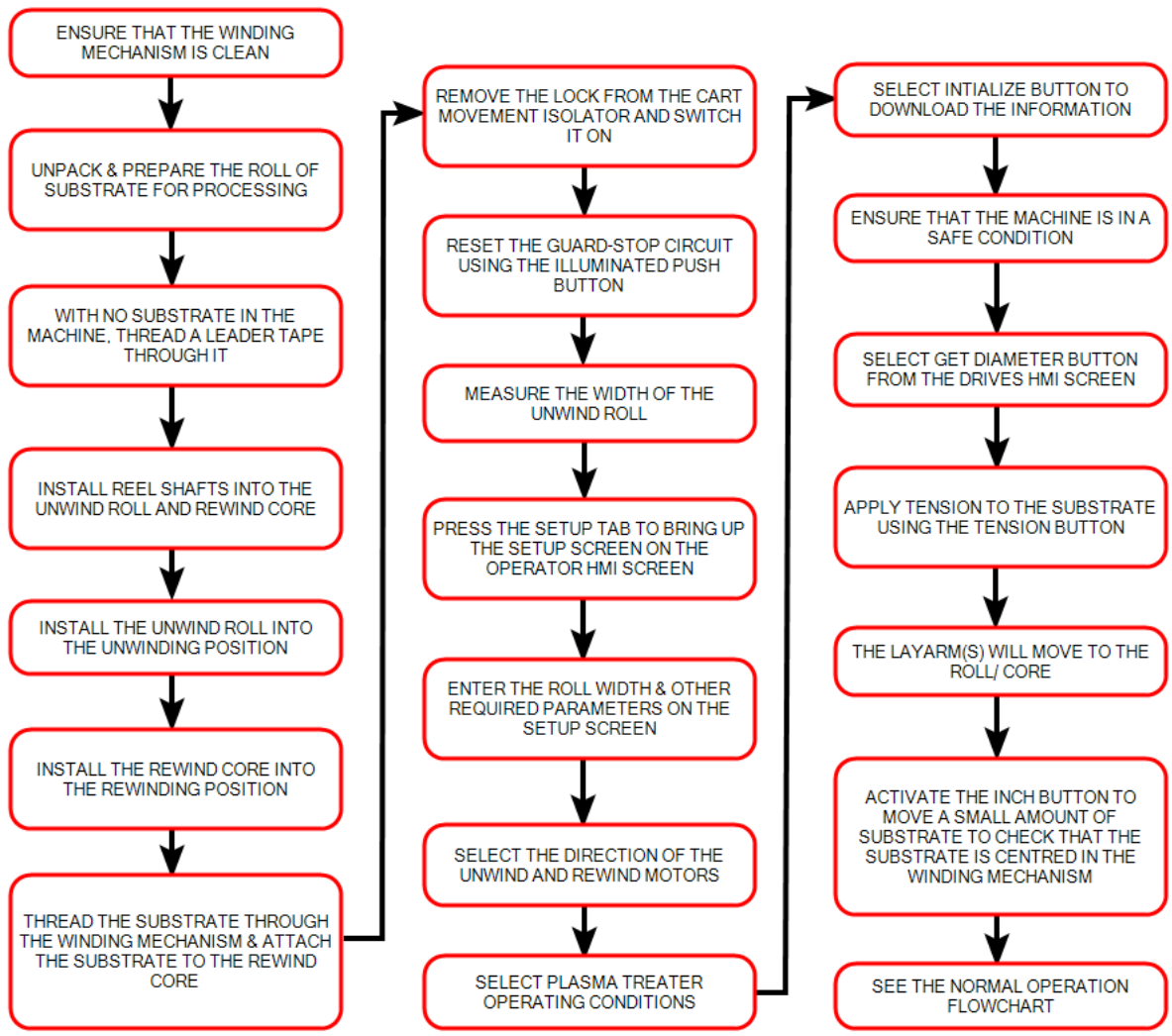


Figure 7.4-2 Preparing the Winding Mechanism for Process Running

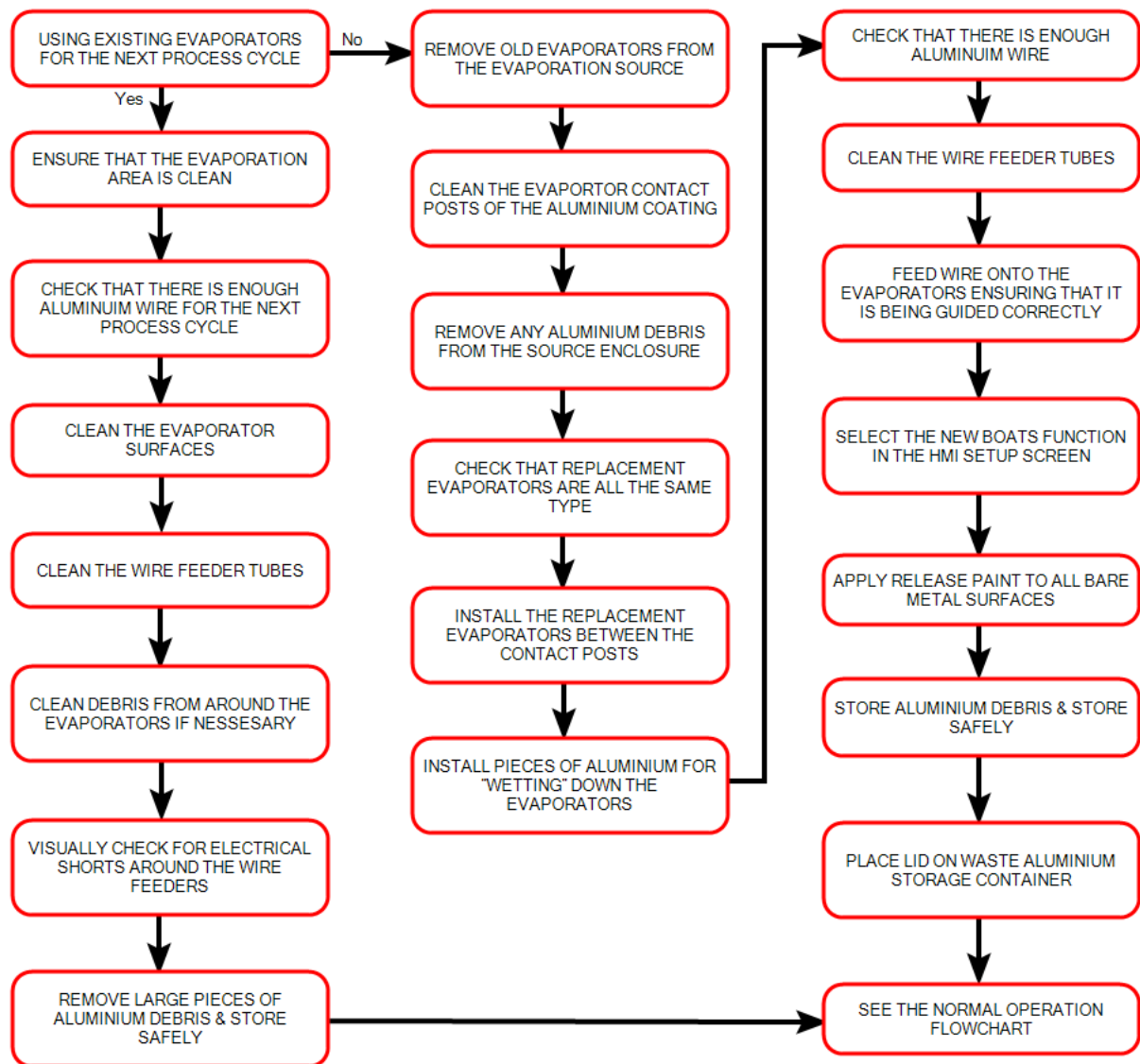


Figure 7.4-3 Preparing the Evaporation Area for Process Running

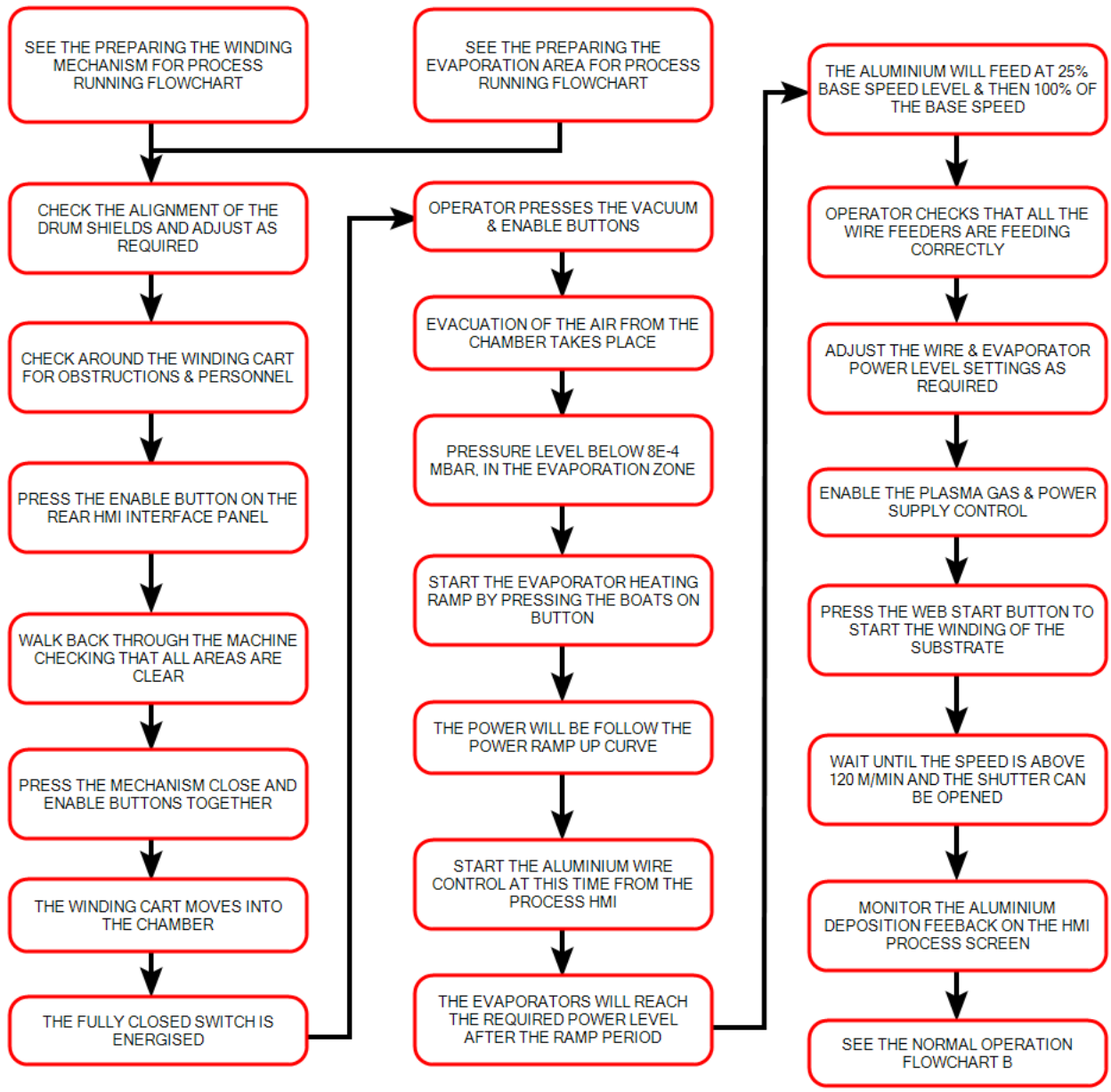


Figure 7.4-4 Normal Operation Flow Chart A

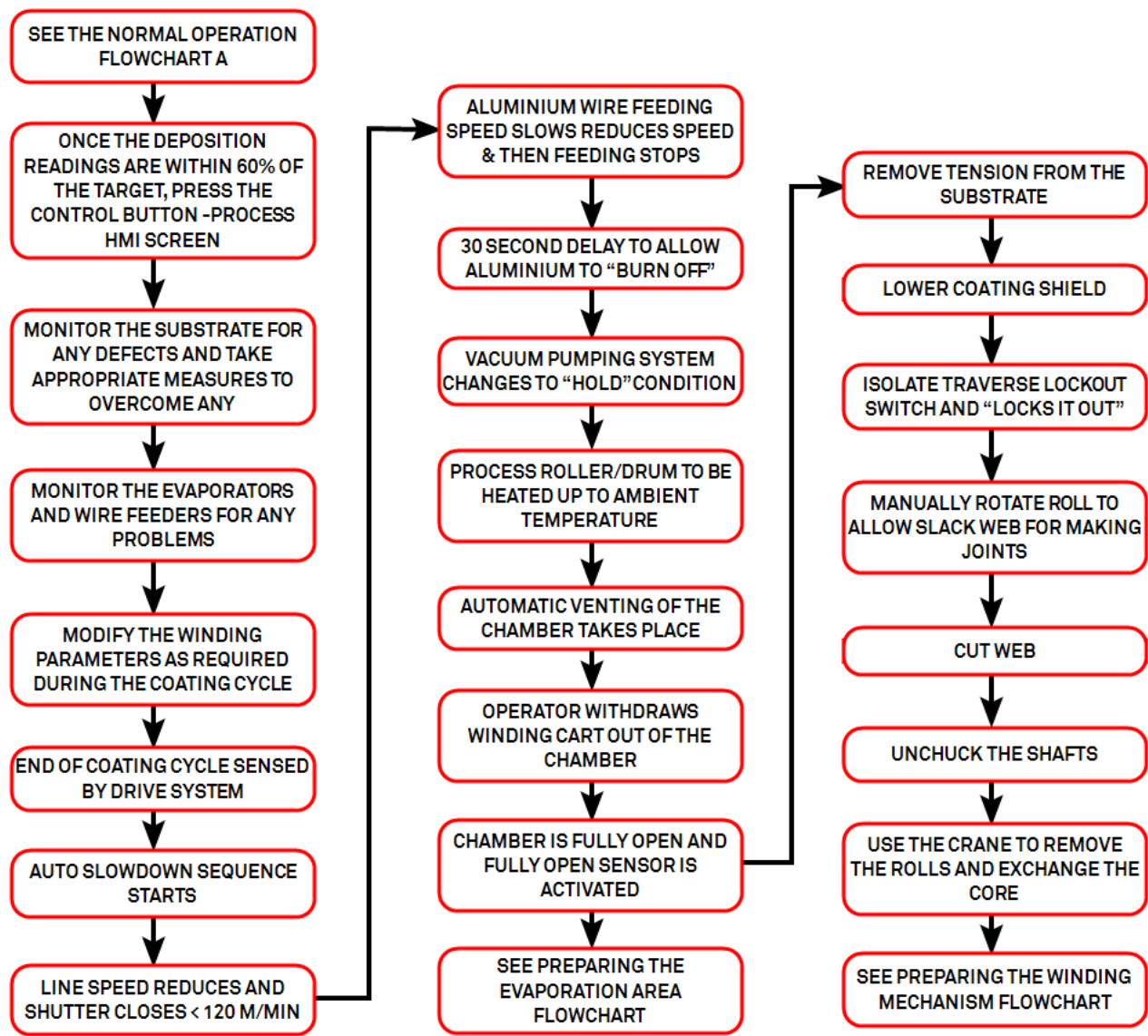


Figure 7.4-5 Normal Operation Flow Chart B

7.5 Operational Sequence of Standard Alubond Coating Process

Rather than trying to explain the sequence of operating the machine for this process with in-depth written descriptions, flow charts are provided, to help the operator of the machine follow the correct operational sequences.

It has been found that operators of this type of equipment prefer this type of information, while operating the machine.

At relevant sections throughout the charts, reference is made to other sections in this manual (section numbers), where further information is provided about any relevant subject.

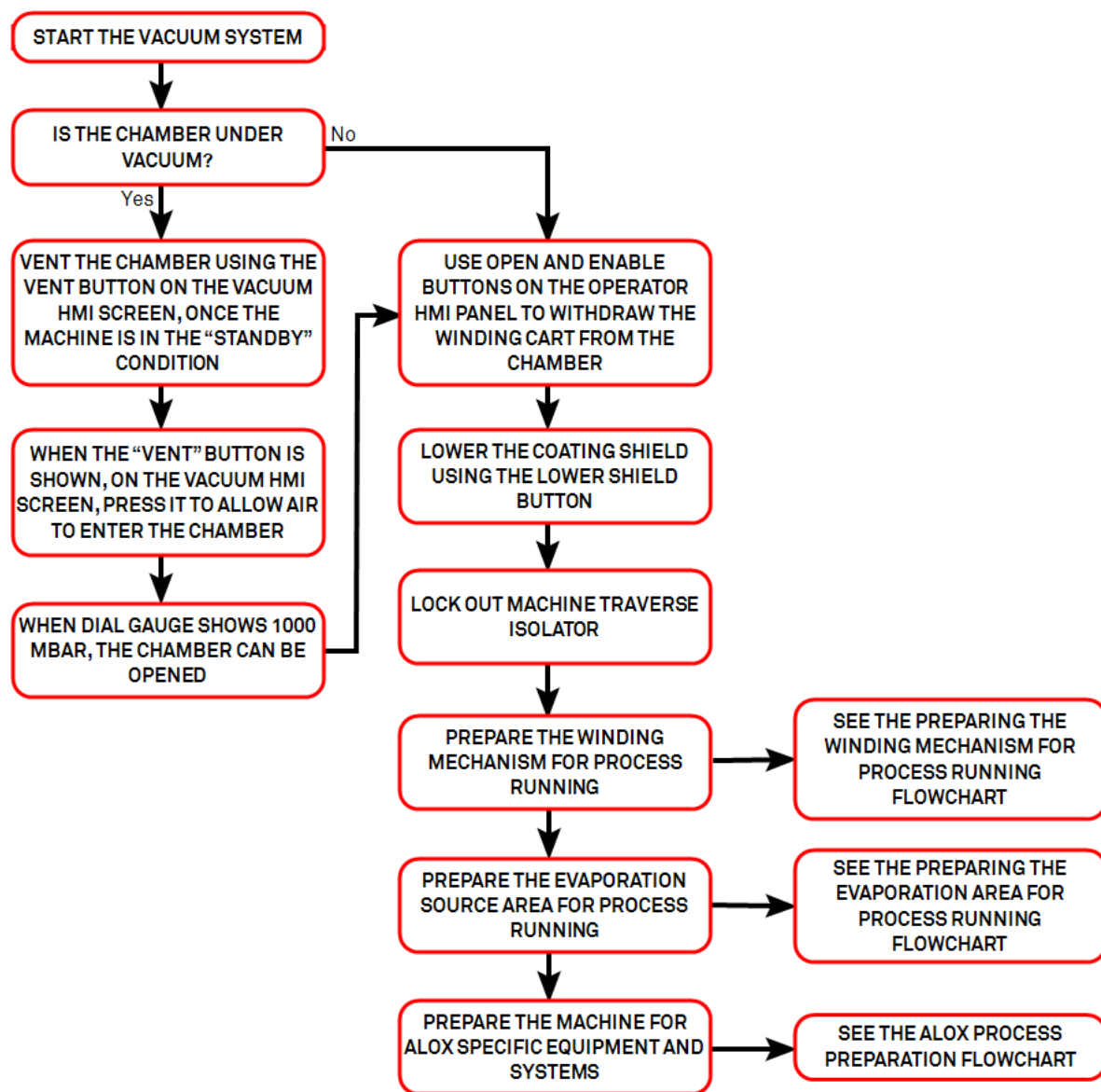


Figure 7.5-1 Machine Operation Flow Chart

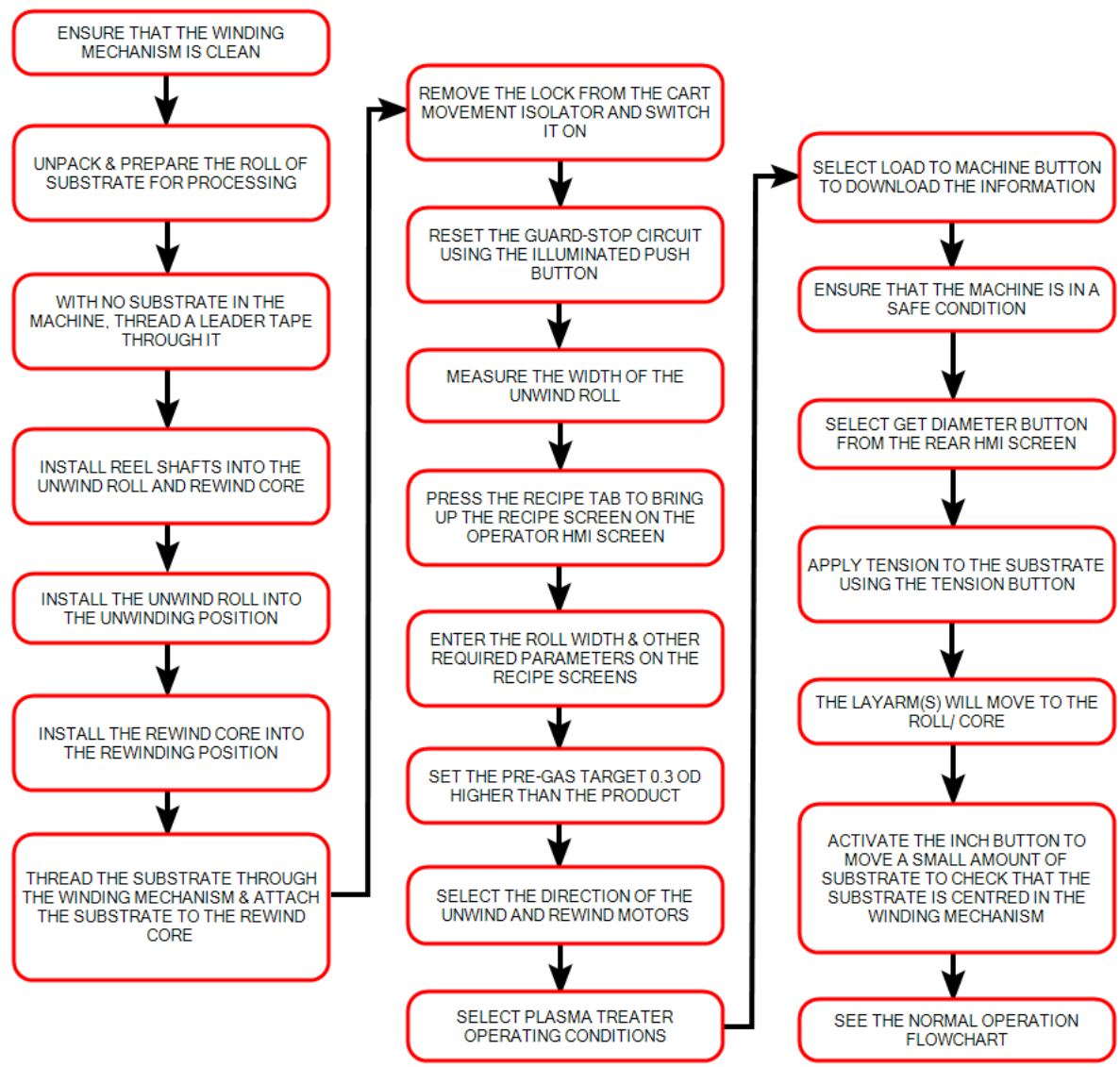


Figure 7.5-2 Machine Operation Flow Chart - Preparing the Winding Mechanism for Process Running

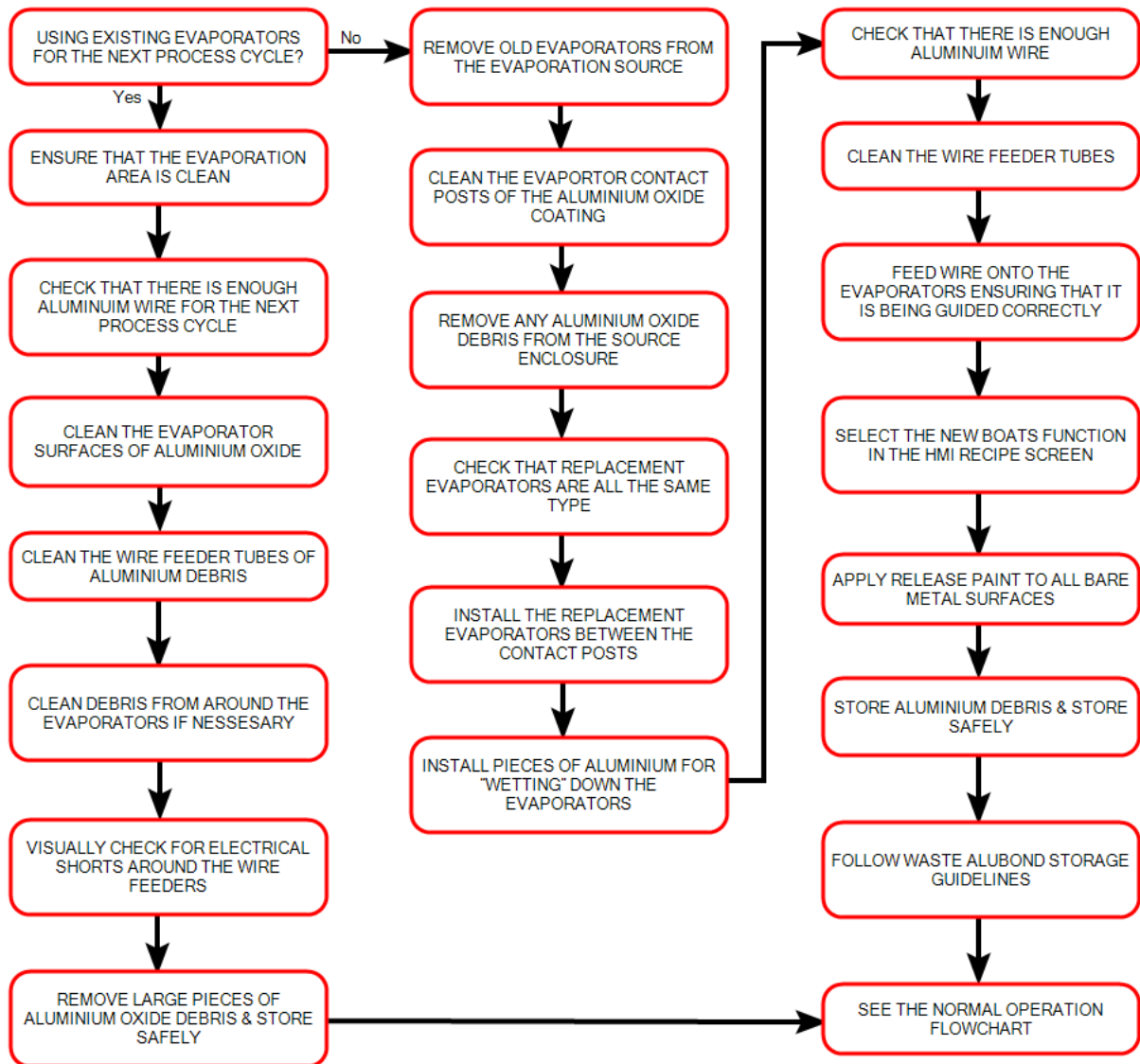


Figure 7.5-3 Machine Operation Flow Chart - Preparing the Evaporation Area for Process Running

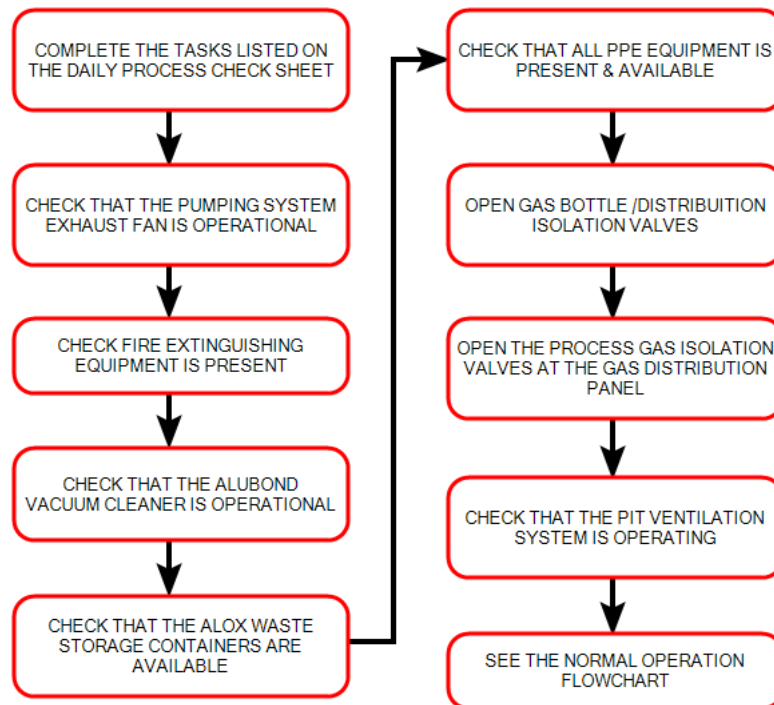


Figure 7.5-4 ALUBOND Process Preparation Flow Chart

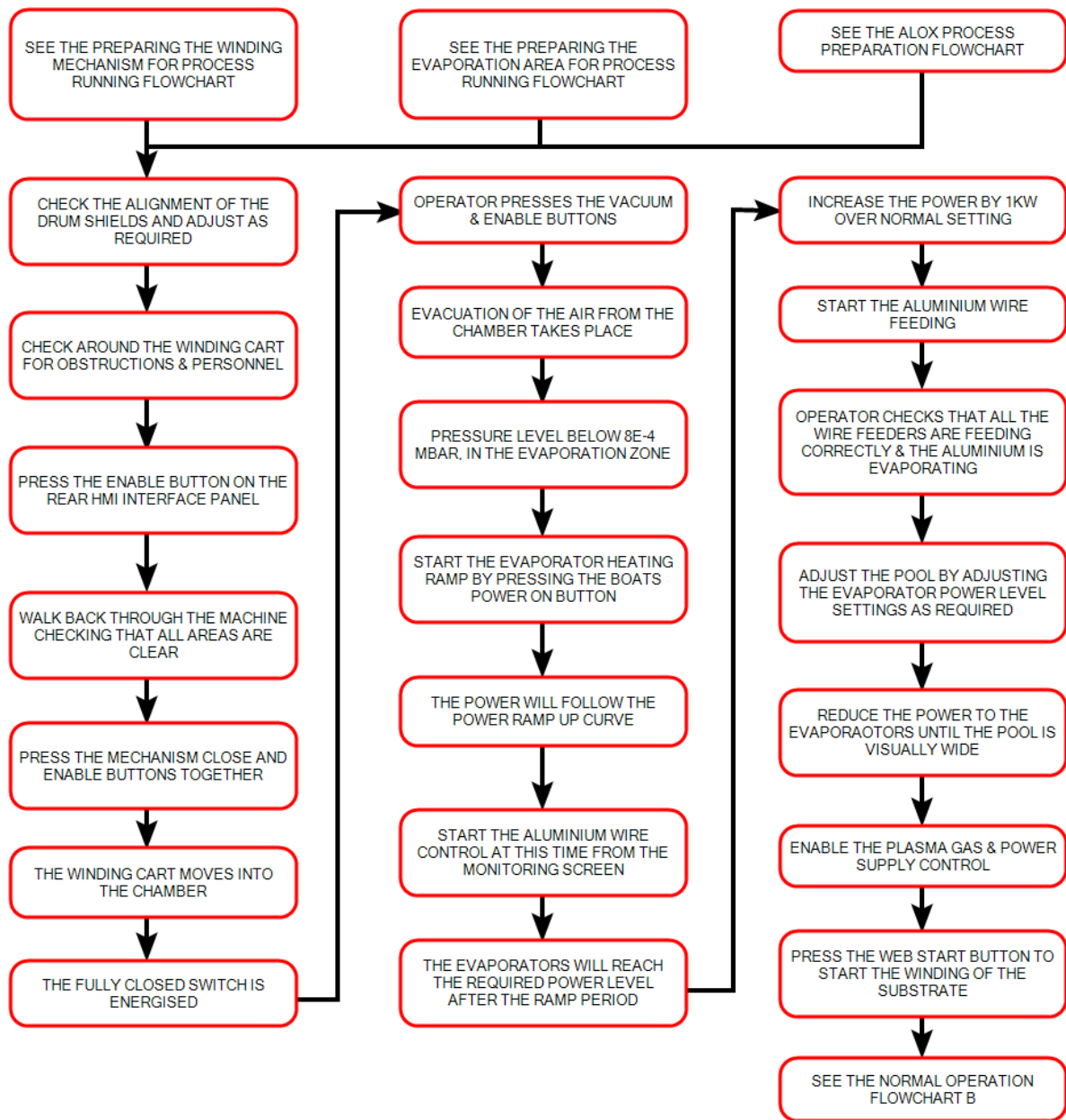


Figure 7.5-5 Normal Operation Flow Chart A

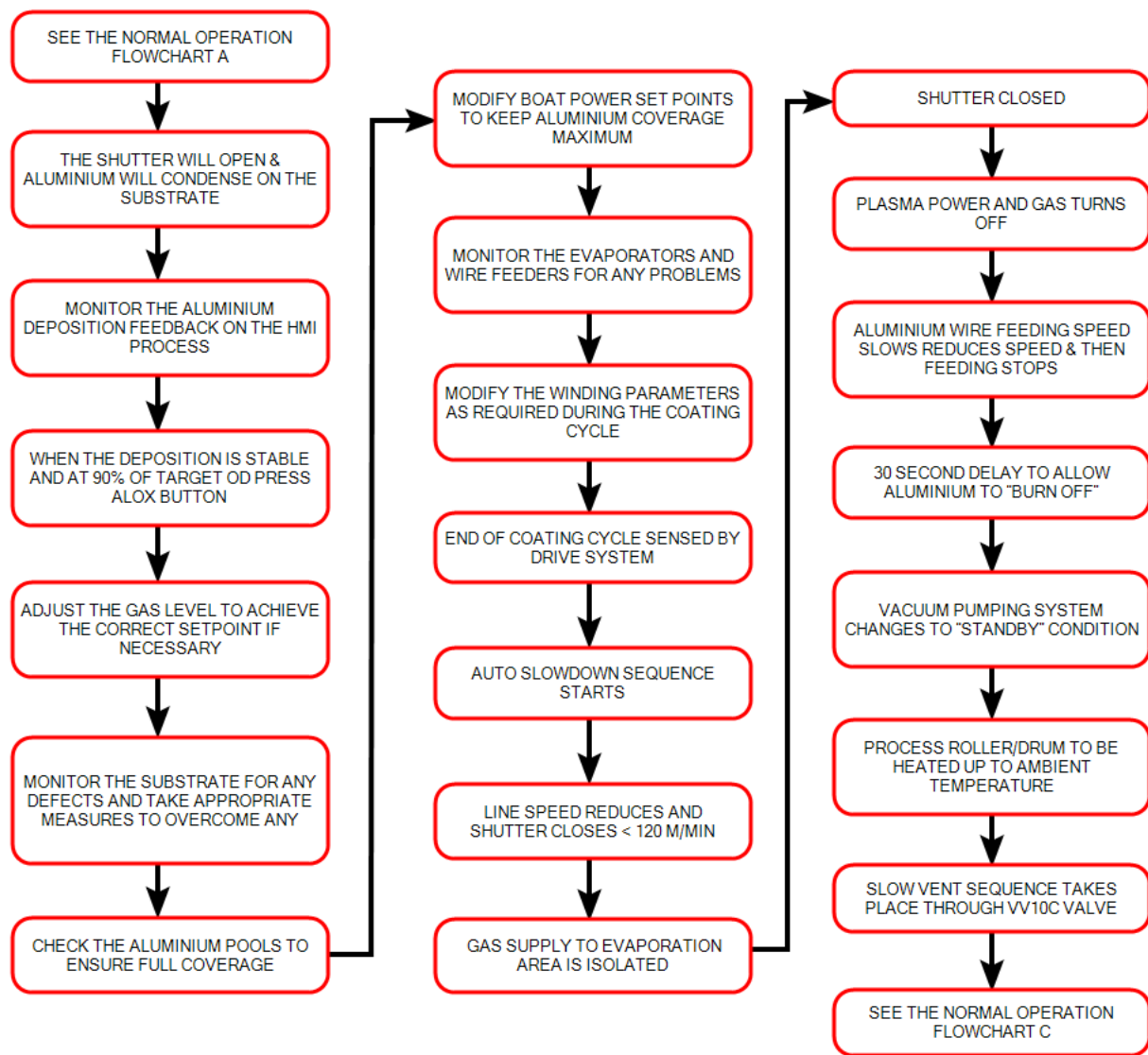


Figure 7.5-6 Normal Operation Flow Chart B

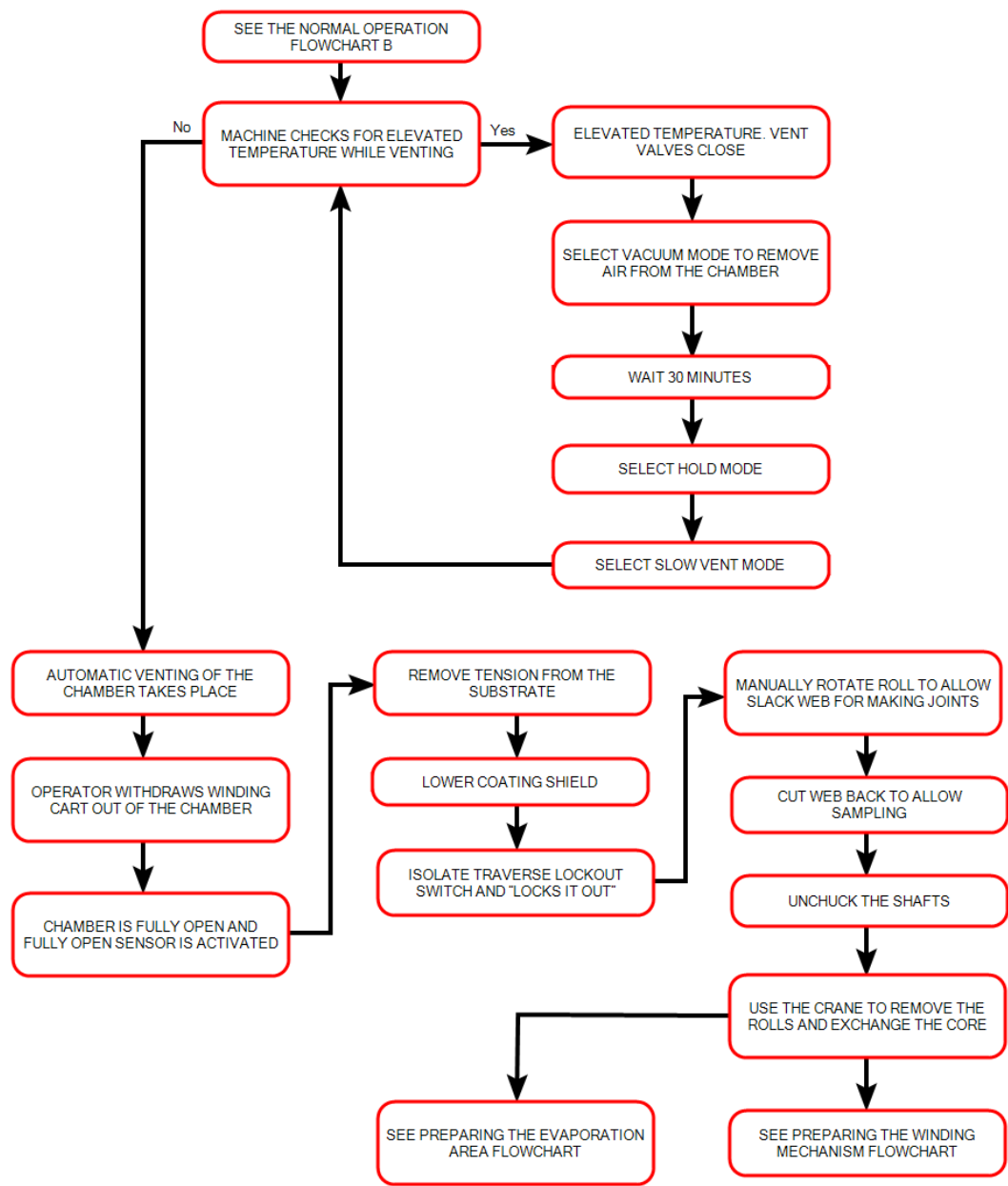


Figure 7.5-7 Normal Operation Flow Chart C

The following chart is provided to allow the user to know what steps have to be carried out when changing between Aluminium, Alubond and ALUBOND process.

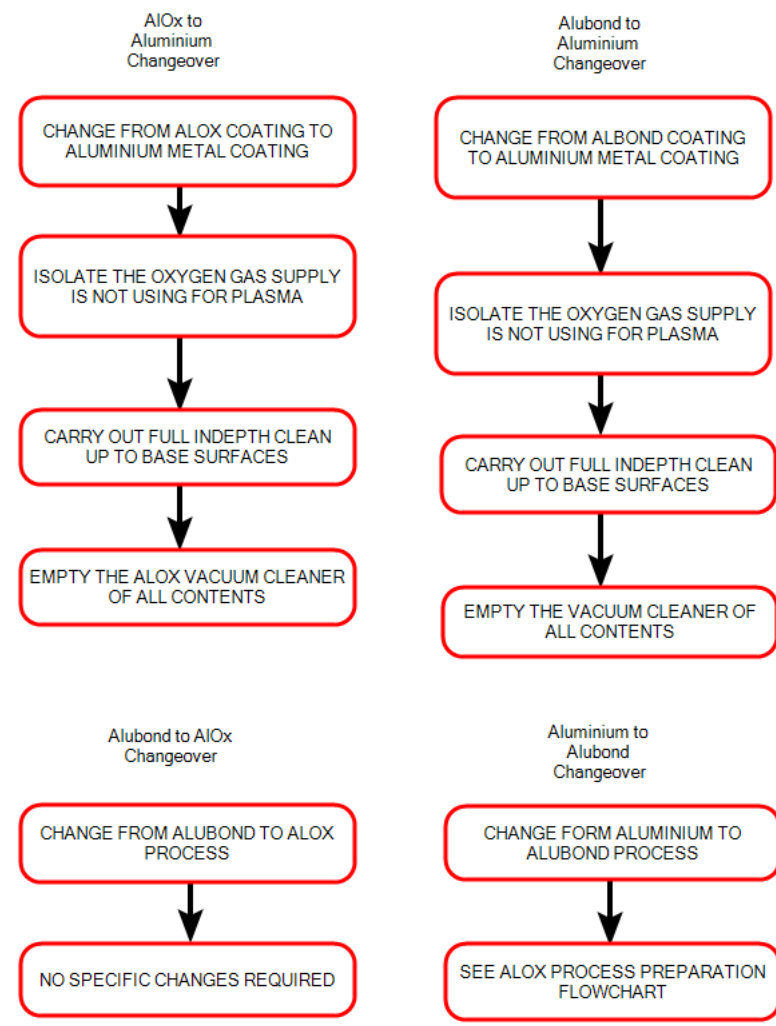


Figure 7.5-8 Process Changeover Flow Chart

The following section highlights tasks that need to be carried out at the end of an ALUBOND process “campaign”

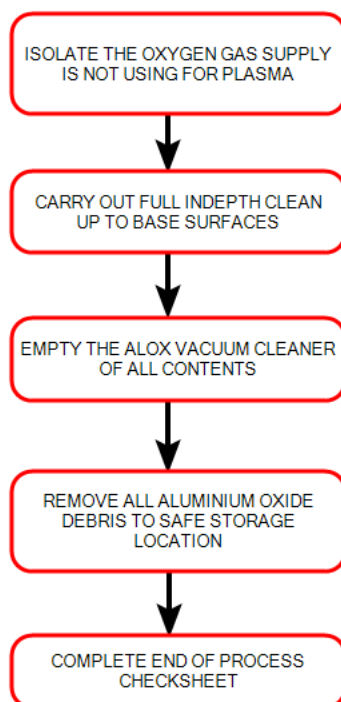


Figure 7.5-9 “End of ALUBOND Process Runs” Flow Chart

7.6 Machine Shutdown Sequence

The machine should be placed in a safe condition.

The vacuum pumping system can be switched off in a correct sequence by pressing the shutdown button on the “Vacuum” HMI screen. The flow diagram for this sequence can be seen below;-

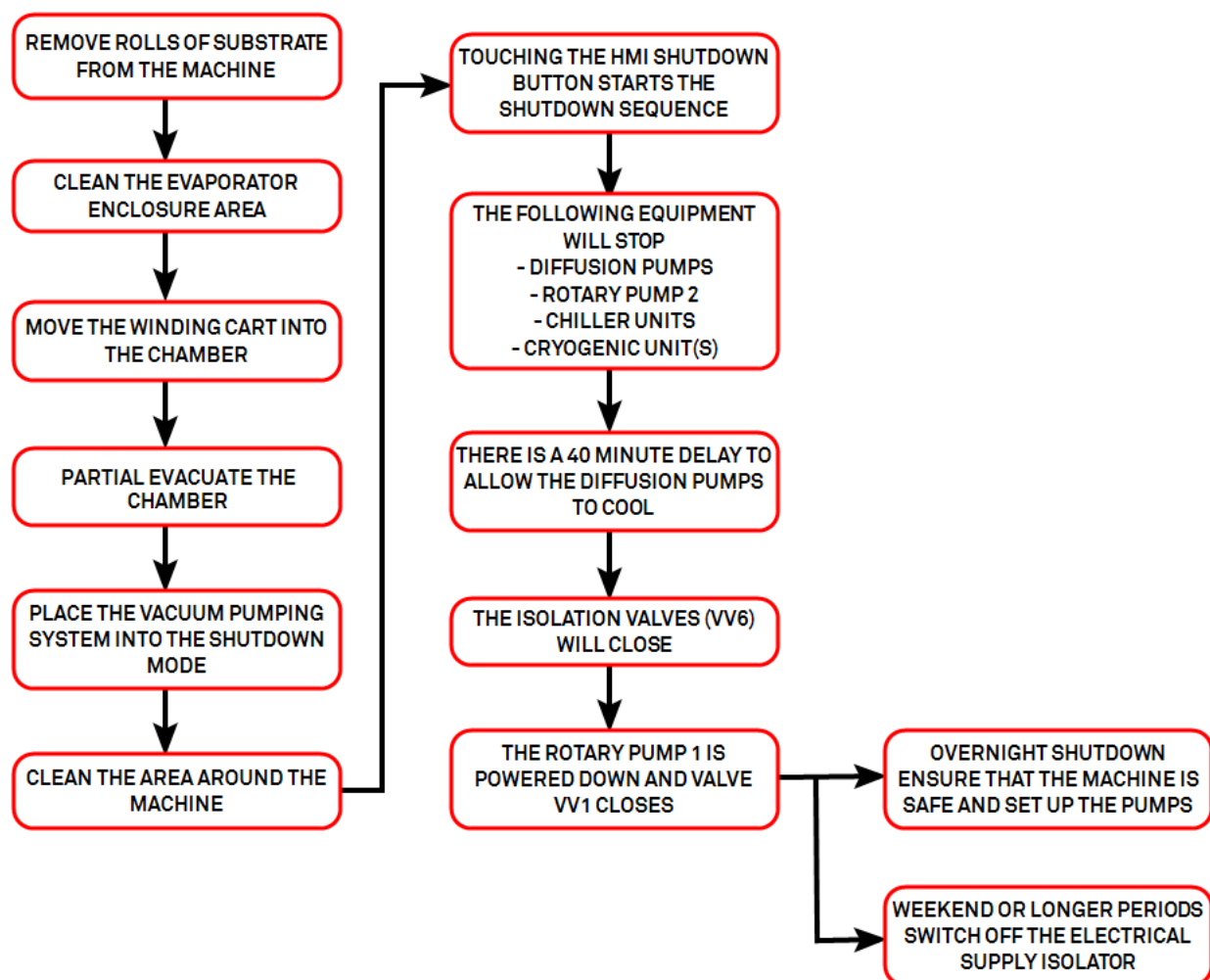


Figure 7.6-1 Shutdown sequence

Partial Evacuation

Before placing the machine in the SHUTDOWN mode.

To keep the chamber clean, it is normal to close the chamber, without any substrate in it and partially pump it down before placing the machine in the shutdown mode. This will ensure that the chamber is kept clean.

It is suggested that the chamber be pumped down to about 500 mbar, and then placed into the HOLD mode.

7.7 Actions by the Operator and Machine When a Web Break Sensed

The machine should be placed in a safe condition, following a “web Break” situation

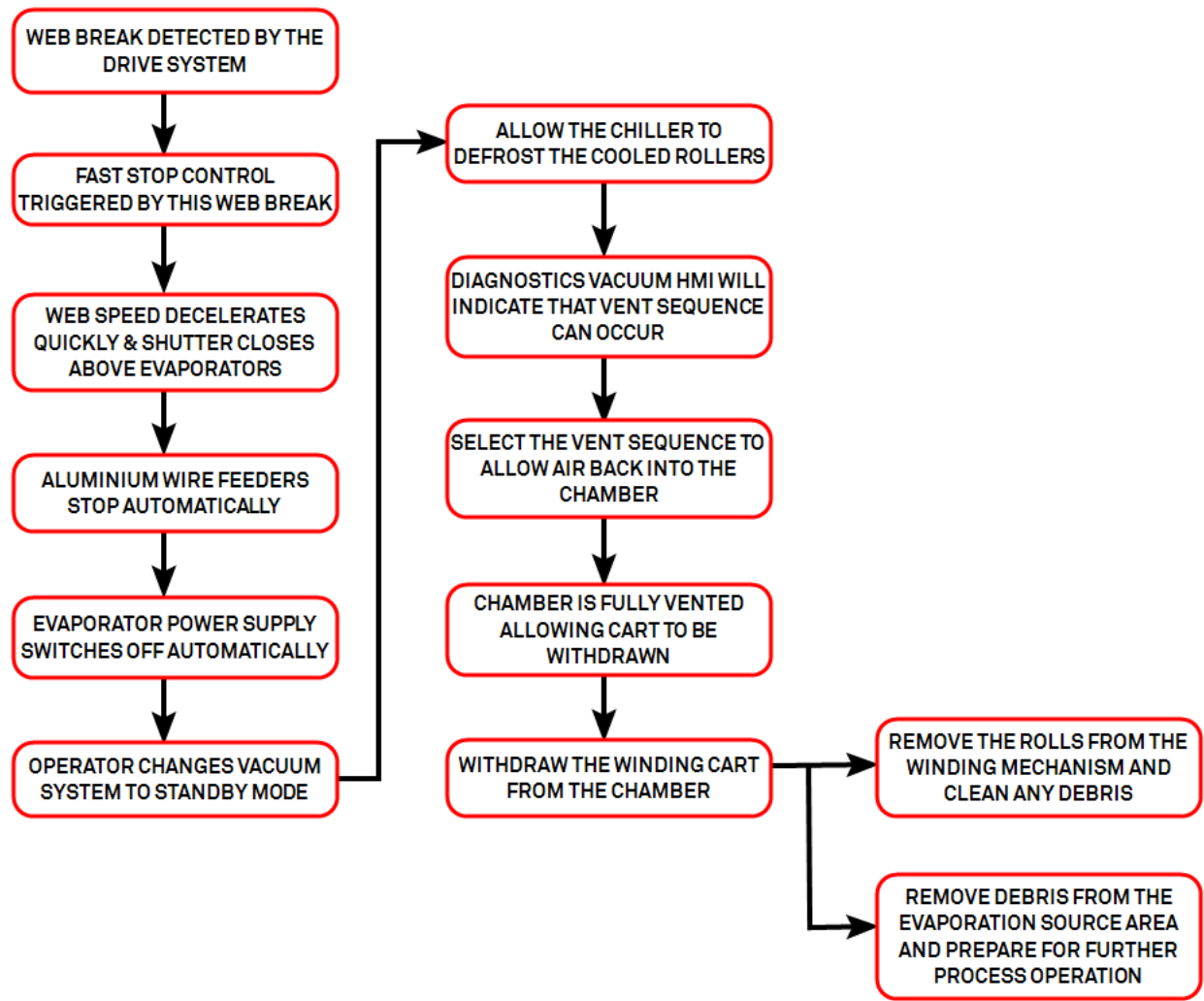


Figure 7.7-1 Action by the Operator and Machine When a Web Break Sensed

7.8 Action by the Operator & Machine when an E-Stop has been selected

The machine should be placed in a safe condition, following a “web Break” situation

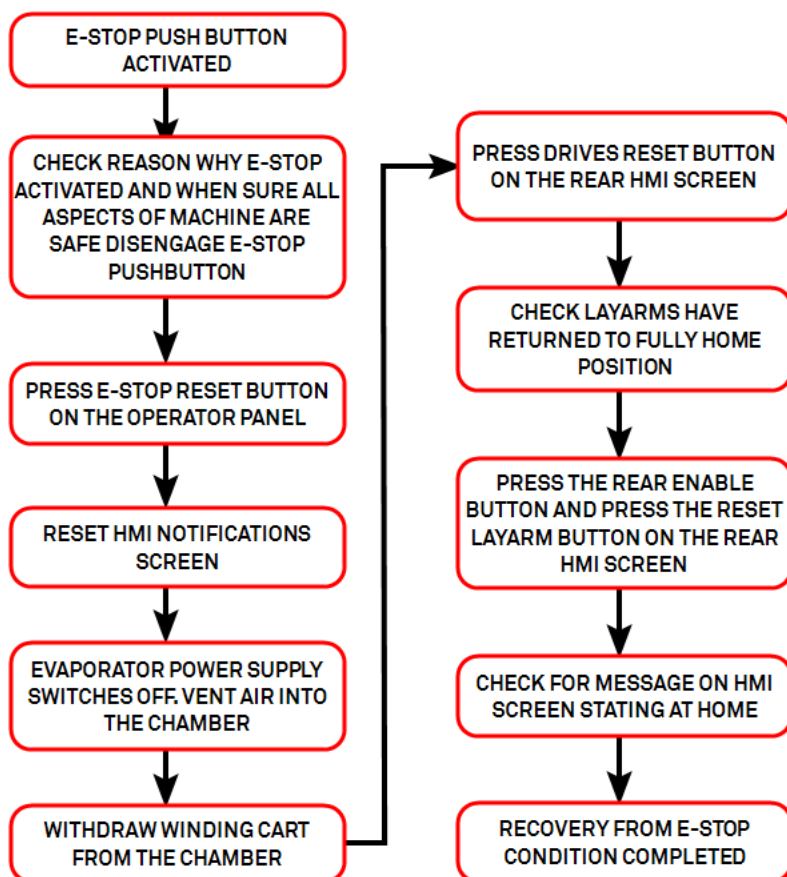


Figure 7.8-1 Action by the Operator and Machine When an E-Stop has been selected

7.9 Actions Required By Operator When Checking Operation of Plasma Treater in Test Mode

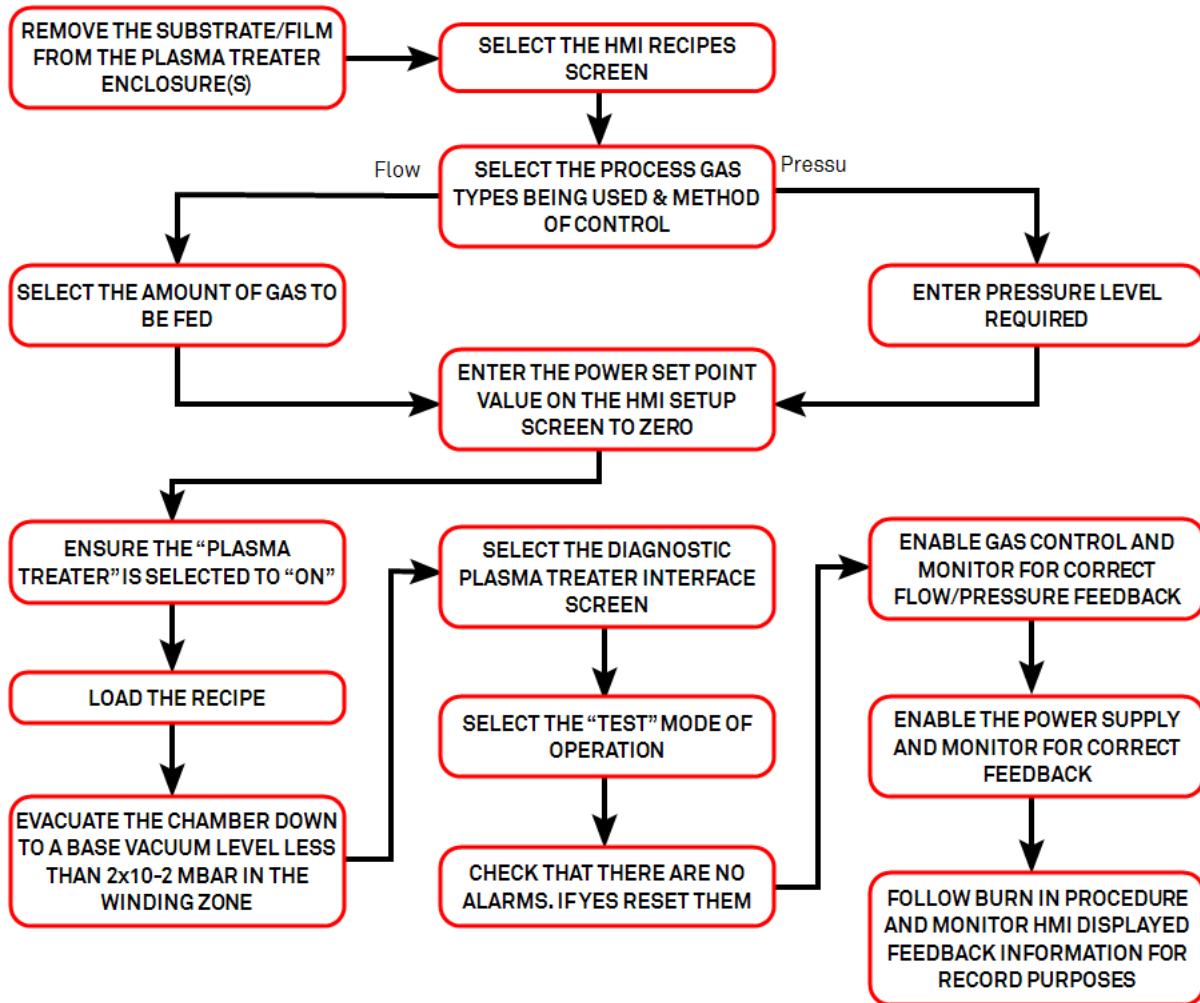


Figure 7.9-1 Actions required by operator when checking operation of plasma treater in test mode

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SECTION 8.0 MACHINE CLEANING PROCESSES

This section describes the machine cleaning processes in the following chapters:

- [8.1 Machine Cleaning on page 349](#)
- [8.2 Cleaning Schedule on page 364](#)
- [8.3 Disposal of Waste Materials on page 366](#)
- [8.4 Process Drum Cleaning on page 368](#)

8.1 Machine Cleaning

A waste product of the aluminium metallizing process is the aluminium that does not deposit on the substrate being coated.

Regular cleaning of the machine is essential if the machine is to maintain a high standard of finished product, good vacuum performance and optimum processing speed.

In addition, if deposited airborne dust is disturbed and falls onto the substrate during the operation of machine, it will cause pinholes in the coating.

Personal Protective Equipment (PPE)

It is advised that operators use personal protection equipment in accordance with the standards specified in Appendix E of this manual and follow the guidance provided by suppliers of any cleaning substances. The following table details the parts of the body at risk during cleaning and recommended clothing to address these risks.

Note!	
	Eyes - Safety goggles (not glasses) (ANSI Z810.1-2003-6.2.1) capable of protecting the user from dust & projectiles should be worn by all personnel on the machine when cleaning is taking place.
	Hands – Gloves capable of protecting the user from abrasion, temperature extremes, cuts and punctures, impact, chemicals.
	Lungs – Respirators (EN 149:2001) capable of protecting the user from breathing in contaminants in the air are required to be worn when cleaning the machine.
	Body – Flame-retardant overalls or boiler suits (ISO 14116:2008) capable of protecting the user from heat and potential fire.
	Feet - Safety boots or shoes with protective toecaps and penetration-resistant, mid-sole capable of protecting the user from sharp or heavy objects.

Operator Hand Tools

It is recommended that good quality-cleaning tools be provided and used in the cleaning procedures described in this section of the manual to provide the most efficient use of the equipment and minimum damage.

		
Blade holders for cleaning process drum surface		
		
Stiff blade scrapers for cleaning surfaces contaminated by aluminium deposits		
		
Special tools to clean hard to access areas		

Figure 8.1-1 A selection of scrapers recommended for cleaning aluminium debris.

	Note !
	The use of inappropriate or damaged tools will lead to machine damage. Do not use steel scrapers or chisels, etc. when cleaning chrome or idler rollers.

Vacuum Cleaner

It is recommended that an industrial rated vacuum cleaner be used to allow a high level of aluminium dust collection from the machine, when cleaning takes place. It should be rated for combustible dust.

If the aluminium dust is not collected, it can become airborne, every time the chamber is vented to atmosphere and contaminate other areas of the environment.



Figure 8.1-2 Example of a suitable industrial type vacuum cleaner

A suggestion of a suitable industrial vacuum supplier is <http://www.nilfiskcfm.com>.

	Warning ! Vacuum cleaners used for aluminium debris should not be used for other material and should be permanently marked for this purpose.
---	--

Aluminium Waste Storage Container

The waste from the process, is required to be removed from the machine and removed from site through a recycle company, as it is still of a high grade.

This includes multi-layer pieces and dust



Figure 8.1-3 Example of a suitable storage drum for aluminium debris

	Warning ! Aluminium dust/debris should be stored in sealable steel containers to stop the contents becoming damp prior to disposal offsite
--	---

The steel drums should have clear identification of the contents that are stored within it and the lid should be applied at all times when access is not required.



Figure 8.1-4 Example of appropriate labelling on the steel container



Figure 8.1-5 Example of the standard aluminium waste collected from a machine

There will be aluminium wire left on each of the spool, which can be recycled.



Figure 8.1-6 Example of the aluminium wire waste collected from a machine that can be recycled

	Note !
	The use of wire brushes and pneumatic tools is discouraged, as they create airborne dust particles.

Cleaning Procedures

The cleaning procedures for different features of the machine are described below.

	Warning!
	Make sure that the incoming energy source is switched off and secured by a padlock (lock out tag out system) during cleaning.

Aluminium Source Box, Shutter & Shield

The source box, shutter and coating shield are subject to heavy coatings of aluminium and must be examined and cleaned at regular intervals, normally between cycles

The cleaning of exposed surfaces can be simplified if they are painted with a release agent such as boron nitride or graphite suspension. If the instructions concerning the release paint/agent have been followed, the aluminium deposits will be easily be removed.



Figure 8.1-7 Example of an operator painting the evaporation source area



Figure 8.1-8 Example of an operator cleaning debris from the shutter

	Note !
	Do not use hammers and chisels to remove the aluminium debris as this can lead to "pits" in the source box surface.

The evaporation source enclosure can be cleaned in with flat scrapers of the type described. Remove the larger pieces of aluminium debris by hand and the smaller pieces and dust using an industrial vacuum cleaner.



Figure 8.1-9 Example of an operator cleaning the coating shield

There are a number of ways of cleaning the aluminium coating off the shield surfaces, scraper, etc. If possible remove large pieces of the coating first and then remove the remainder.

Finally clean all areas with an industrial vacuum cleaner.



Figure 8.1-10 Example of an operator using a vacuum cleaner to remove small debris



Figure 8.1-11 Example showing a mat placed on the floor under the coating shield to allow debris collection

Removable Collection Shields

There are a number of removable shields provided around the evaporation enclosure. These are provided to collect the stray aluminium evaporation.

These are manufactured out of aluminium/stainless steel, and can be cleaned using scrapers outside the chamber.

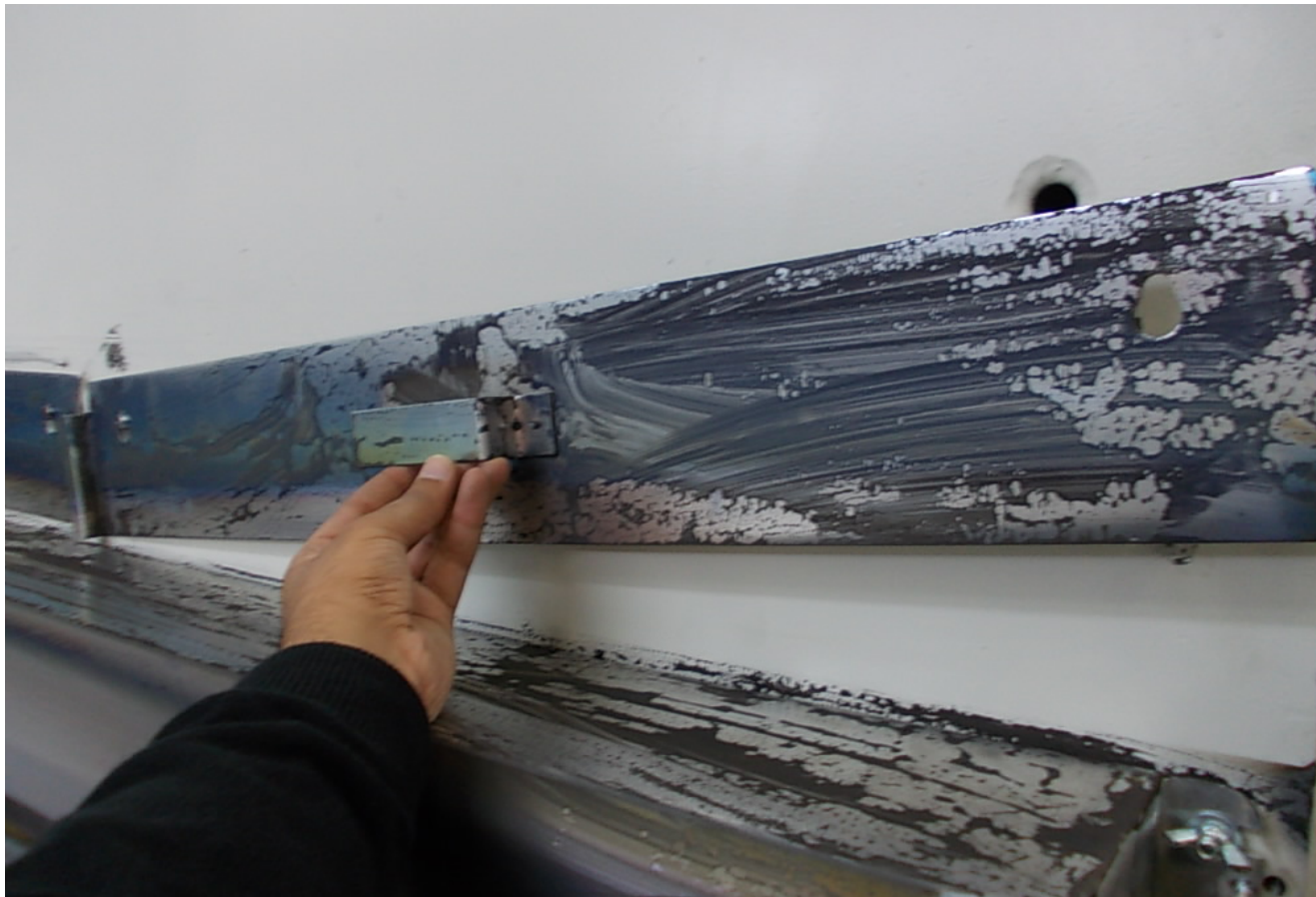


Figure 8.1-12 Example of an operator removing a coating shield

Evaporator Posts/Contact Blocks

To extend the life of the evaporator contact blocks, it is suggested that all debris be removed carefully from them when replacing the evaporators.

If the contact surface is damaged, poor contact between the evaporators and the blocks can take place, which will shorten the useful life of the blocks.

It is suggested that when evaporators are not being used, as the width is narrower, that the non-used evaporators are covered in thick aluminium foil to keep them clean.



Figure 8.1-13 Example showing foil placed over evaporators that are not in use

Cleaning of Wire Feed Spouts

The piece of tube that guides the aluminium wire onto the evaporator is known as a spout. They are located in the vapour cloud and become heavily coated with aluminium debris. If not cleaned on a regular basis, they can become blocked and the aluminium wire will not feed through them.

	Note !
	Care must be taken not to damage the tubes by crushing the tubes when cleaning them.

Channel lock 748 8" End Cutting Long Reach Pliers will not crush the tube, as there are stops at their end.

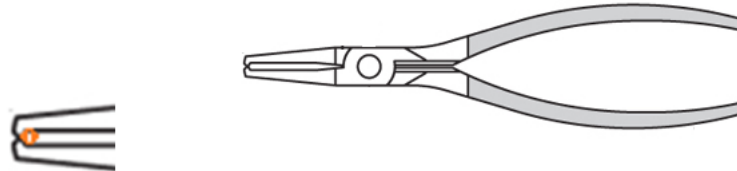


Figure 8.1-14 End cutting long reach pliers



Figure 8.1-15 Example of an operator using long pliers to clean the wire feed tube

The spouts are classified as a “consumable” item and it is expected that they will require replacing on a regular basis.



Figure 8.1-16 Examples of an operator cleaning the wire feeder tubes

The aluminium debris can be removed by twisting its outer surface lightly with a pair of pliers, taking care not to damage the spout. They should be painted with release agent after cleaning.

If required, to clean the inner part of the tube, they can be placed in a weak caustic solution, e.g. commercial drain cleaner. Follow all the highlighted precautions as instructed on the cleaner's container.

It is suggested that aluminium foil be placed around the clamp section of the feed tube to keep it clean.



Figure 8.1-17 Operator putting foil on the wire feeder clamps

Winding Mechanism Rollers

	Warning ! Rollers must only be rotated by hand while cleaning takes place and the power should be switched off and locked out. Only on machines fitted with a drum, cleaner pedal is cleaning the drum permitted under power.
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Rubber Cut Spreader Rollers

	Note ! Do not use any gasoline derivatives, MEK, Acetone and ethyl acetate as these will attack and swell the rubber. Avoid the use of nylon or stiff brushes as damage may occur to the roller surface.
---	--

The manufacturer recommends that the sleeve be cleaned with water or a cloth dampened with Ultrasol 550. Do not use an excessive amount of solvent. Bobst Manchester Ltd. can provide data Sheet. Any light pure hydrocarbon based solvent that is low in aromatic content is good - for example "naphtha".

Ultrasol 550 can be purchased from-

E P Laboratories Ltd

Orchard House Industrial Estate

Amersham Road

Chesham

HP5 1NE

England

Tel: +44 (0) 1494 791585

Fax: +44 (0) 1494 771853

E-mail: info@eplabs.com

Website is www.eplabs.com

Kickert Rubber Sleeved Bow Type Spreader Rollers

No aggressive solutions or sharp tools may be used because they will damage the rubber sleeve.

The manufacturer recommends that the outer rubber sleeve be cleaned with a cloth dampened with Akemina NITRO-dilution.

Cork Covered Rollers

Use an adhesive backed tape to lift dust from the surface of the roller. The cork cannot be cleaned perfectly and will eventually need to be replaced. Use adhesive backed cork for this purpose. Bobst Manchester Ltd. stocks this material.

Do not attempt to use cork without adhesive backing. It has been found that using a non-backed cork and glue causes contamination problems and unevenness of the surface.

Chrome Idler Rollers & Process Drum

Clean the chrome surface of the roller with a metal polish. The residue should be removed with a clean cloth before wiping the whole surface with a cloth damped with "Acetone".

In the case where the drum has been coated with a light to medium coating of aluminium, in cases, for example, where a web break has occurred, use a scraper of the type shown below. The cleaning motion should be in the direction around the drum, not across its surface.



Figure 8.1-18 An operator cleaning the drum



Figure 8.1-19 recommended tools for use in cleaning deposits from the process drum.

	Note !
	Care must be taken not to damage the surface of the drum whilst cleaning with a blade.

Cleaning of Contaminated Bearings

Some bearings on the machine are not fully sealed, such as those used in the load cell transducers, and contamination can occur.

If the winding area of the machine is kept clean of dust, excessive grease can be removed by wiping the outside of the bearing with a cloth. However, to clean the bearings of lubricating grease and aluminium debris, the bearings should be removed from the machine.

The suggested method of cleaning off the machine is -

1. Remove the bearings from the machine;
2. Clean in a degreasing bath or cleaner using degreasing solvents;
3. Once the excess grease and debris have been removed, submerge the in a container of acetone to remove any remaining solvents used during the degreasing stage;
4. Remove and place in a further separate container containing acetone before greasing with the recommended grease and reinstalling on the machine.

**Caution !**

When using new replacement bearings, they have to be fully degreased and then refilled with the appropriate vacuum-rated grease. Failure to complete this procedure may lead to the original lubricant outgassing into the vacuum chamber.

Cleaning of Layarm slides

The layarm assemblies slide on linear bearings that travel along the slide rods. Over a period, debris can build up on the rods. It is suggested that these are cleaned by wiping them with a lint free cloth damped with lightweight oil (use the same oil as in the gravity-filled bottles on the rotary leadthroughs).

It may become necessary to remove the assembly to replace the bearings or the rod if it becomes damaged and pitted. Please contact BOBST MANCHESTER LTD. for all machine spares.

**Caution !**

Do not use release agents such as WD40, as they potentially have a high out-gassing rate.

Cleaning Coating Shield Concave Zoning Seals

The concave aluminium zoning strips provide a low conductance seal between the winding and evaporation zones of the chamber.

To remove the debris, it is suggested that a scraper be used to remove heavy deposits, while light build-ups, can be removed with a slightly abrasive cloth.

Cleaning of Flexible Zoning Seals

The suggested method of cleaning is to use a slightly damp cloth to wipe away any debris from the seal. The seal should be painted with release paint/agent to allow the mating surfaces to engage without binding.



Figure 8.1-20 Example of an operator inspecting the flexible zone seal.

Vessel Walls

If the machine is, being operated correctly the remaining stray aluminium deposits on the vessel wall and mechanisms will be small but should be cleaned once a week to prevent a build-up.

The main vessel flange and seals should be examined on a regular basis and kept clean.

Care should be taken not to inadvertently introduce any high vapour pressure substances into the vessel during cleaning, e.g. certain solvents.

Internal Viewing Windows

Use a dry cloth to remove all dust that has built up on the window.